

Theorem Let $\mathbb{F}$: field, $f \in \mathbb{F}[X_1, \dots, X_n]^{S_n}$. $\exists! g \in \mathbb{F}[X_1, \dots, X_n]$ s.t. $f = g(e_1, \dots, e_n)$ where e_i are elementary symmetric polynomials.

In fact, this works if $\mathbb{F}$ is only an integral domain by $R \hookrightarrow \text{Frac}(R)$: field.

Examples (Discriminant) $f = X^n - c_1 X^{n-1} + \dots + (-1)^n c_n = \prod_{i=1}^n (X - \alpha_i)$. Define

$$\text{disc}(f) := \prod_{i < j} (\alpha_i - \alpha_j)^2.$$

If we view $\alpha_1, \dots, \alpha_n$ as variables, then $\text{disc}(f)$ is a symmetric polynomial in $\alpha_1, \dots, \alpha_n$.

$\because \forall \sigma \in S_n, \sigma$ sends $\prod_{i < j} (\alpha_i - \alpha_j)$ to $\pm \prod_{i < j} (\alpha_i - \alpha_j)$.

$\xRightarrow{\text{Thm}}$ $\text{disc}(f)$ is a polynomial (uniquely) in $e_i(\alpha_1, \dots, \alpha_n) = c_i, 1 \leq i \leq n$. This works on $\mathbb{Q}$ (and $\mathbb{Z}$ if we accept that Thm works on integral domains).

Examples

- $f = X^2 - bX + c = (X - \alpha_1)(X - \alpha_2)$,
 $\text{disc}(f) = (\alpha_1 - \alpha_2)^2 = (\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2 = b^2 - 4ac$.
- $f = X^3 + pX + q = (X - \alpha_1)(X - \alpha_2)(X - \alpha_3)$,
 $\text{disc}(f) = -4p^3 - 27q^2$.

§ Resultant

Motivation Let $\mathbb{F}$: field, $f, g \in \mathbb{F}[X]$. To determine whether f, g are coprime, the classical solution is Euclidean algorithm. But we also want an “equational” criterion, i.e. some function whose result determines directly whether f, g are coprime.

Definition Fix $n, m \in \mathbb{Z}_{\geq 1}$, consider $f = v_0 X^n + \dots + v_n, g = w_0 X^n + \dots + w_m, v_i, w_j \in \mathbb{F}$.

Resultant of f and g is defined as

$$\text{Res}(f, g) := \begin{vmatrix} v_0 & \dots & \dots & v_n & & & \\ & \ddots & & & \ddots & & \\ & & v_0 & \dots & \dots & v_n & \\ w_0 & \dots & \dots & w_m & & & \\ & \ddots & & & \ddots & & \\ & & \ddots & & & \ddots & \\ & & & w_0 & \dots & \dots & w_m \end{vmatrix}.$$

Properties

- View v_i, w_j as variables, then $\text{Res}(f, g)$ is a polynomial/ $\mathbb{Z}$ in $v_0, \dots, w_m$ independent of $\mathbb{F}$.
- The coefficient of $v_0^m w_m^n$ is 1, $\because$ only contribution comes from diagonal line.
- $\text{Res}(f, g) = (-1)^{nm} \text{Res}(g, f)$ by swapping rows.
- $\forall t \in \mathbb{F}, \text{Res}(tf, g) = t^m \text{Res}(f, g), \text{Res}(f, tg) = t^n \text{Res}(f, g)$.

Lemma Given n, m and f, g , then

$$\text{Res}(f, g) = 0 \iff \exists f_1, g_1 \in \mathbb{F}[X] \text{ s.t. } fg_1 + gf_1 = 0,$$

where f_1, g_1 not both = 0, $\deg f_1 < n, \deg g_1 < m$.

Proof Write $f_1 = a_1 X^{n-1} + \dots + a_n, g_1 = b_1 X^{n-1} + \dots + b_m, a_i, b_j \in \mathbb{F}$.

$$fg_1 + gf_1 = 0 \iff \begin{cases} b_1 v_0 + a_1 w_0 = 0 \\ b_1 v_1 + b_2 v_0 + a_1 w_1 + a_2 w_0 = 0 \\ \dots \\ b_m v_n + a_n w_m = 0. \end{cases}$$

This homogeneous system of linear equations is equivalent to the transpose of $\text{Res}(f, g)$, which have the same determinant. The existence of $f_1, g_1 \iff \det = 0$ by Cramer's rule. $\square$

Theorem Given n, m and f, g as before,

$$\text{Res}(f, g) = 0 \iff \begin{cases} v_0 = 0 = w_0 \\ \text{or } \exists h \in \mathbb{F}[X], \deg h > 0, h|f \wedge h|g. \end{cases}$$

Proof ($\Leftarrow$) If $v_0 = 0 = w_0$, then the first column of $\text{Res}(f, g)$ is all 0 $\implies \det = 0$.

If $\exists$ such h , then $f \cdot \underbrace{\frac{g}{h}}_{\deg < m} + g \cdot \underbrace{\frac{-f}{h}}_{\deg < n} = 0 \xrightarrow{\text{Lem}} \text{Res}(f, g) = 0$.

($\implies$) Assume $\text{Res}(f, g) = 0$.

- One of f and g is zero, say $f = 0$.
 - $\deg g > 0 \implies$ take $h = g$.
 - $g \in \mathbb{F} \implies$ get $w_0 = 0$ since $m \geq 1$.
- f, g are both nonzero. May assume $(v_0, w_0) \neq (0, 0)$, say $w_0 \neq 0$.

$\xrightarrow{\text{Lem}} \exists f_1, g_1$ not both zero, $f_1 g + g f_1 = 0, \deg g_1 < m$.

In $F(X) := \{\text{rational functions}/\mathbb{F}\} = \text{Frac}(\mathbb{F}[X])$, $\frac{f}{g} = \frac{-f_1}{g_1}$. If f is coprime with g , we would get $g \mid g_1$, contradicting $\deg g_1 < m = \deg g$. $\square$

Theorem Fix $n, m \in \mathbb{Z}_{\geq 1}$, $f = a \prod_{i=1}^n (X - \alpha_i), g = b \prod_{j=1}^m (X - \beta_j)$ with all variables $\in \mathbb{F}$, then

$$\begin{aligned} \text{Res}(f, g) &= a^m \prod_{i=1}^n g(\alpha_i) = (-1)^{nm} b^n \prod_{j=1}^m f(\beta_j) \\ &= a^m b^n \prod_{i,j} (\alpha_i - \beta_j). \end{aligned}$$

Proof (Last equation) $(-1)^{mn} b^n \prod_{j=1}^m f(\beta_j) = (-1)^{mn} a^m b^n \prod_{i,j} (\beta_j - \alpha_i) = a^m b^n \prod_{i,j} (\alpha_i - \beta_j)$.

(Second equation) Use $\text{Res}(f, g) = (-1)^{mn} \text{Res}(g, f)$, reduce to first equation.

To show: $\text{Res}(f, g) = a^m \prod_i g(\alpha_i), \text{Res}(tf, g) = t^m \text{Res}(f, g) \implies$ may assume $a = 1$.

- Special case: $g(\alpha_i)$ are pairwise distinct. Now consider $\text{Res}(f, g - Y) \in \mathbb{F}[Y] \subset F(Y), y = 0$ yields $\text{Res}(f, g)$. Note that the top term is $(-1)^n Y^n$.

Also, f and $g - g(\alpha_i)$ have a common root $\alpha_i \xrightarrow{\text{Thm}} \text{Res}(f, g - g(\alpha_i)) = 0$, i.e.

$$(\text{Res}(f, g - Y))(g(\alpha_i)) = 0 \implies (g(\alpha_i) - Y) \mid \text{Res}(f, g - Y).$$

From assumption we can reach

$$\prod_{i=1}^n (g(\alpha_i) - Y) \mid \text{Res}(f, g - Y),$$

Both sides have $\deg = n$ and leading coefficient $= (-1)^n$, hence they are equal. Set $y = 0$ on LHS to reach the conclusion.

- General case “Perturbation”, or argue as follows:

Keep g , assume $b \neq 0$ and take $\tilde{f} = \prod_{i=1}^n (X - Z_i)$ where Z_i are formal variables. Special case (for $F(Z_1, \dots, Z_n)$) gives

$$\text{Res}(\tilde{f}, g) = \prod_{i=1}^n g(Z_i),$$

since $g(Z_i)$ are pairwise distinct in $\mathbb{F}[Z_1, \dots, Z_n]$.

Put $Z_i \leftarrow \alpha_i$ to get $\text{Res}(f, g) = \prod_{i=1}^n g(\alpha_i)$. □

Back to discriminants, given $f = a \prod_{i=1}^n (X - \alpha_i) \in \mathbb{F}[X]$, set $\text{disc}(f) := a^{2n-2} \prod_{i < j} (\alpha_i - \alpha_j)^2$.

Corollary

• $a \cdot \text{disc}(f) = (-1)^{\frac{n(n-1)}{2}} \text{Res}(f, f')$.

Proof Theorem $\implies \text{Res}(f, f') = a^{n-1} \prod_{i=1}^n f'(\alpha_i)$. From $f' = a \sum_{k=1}^n \prod_{j \neq k} (X - \alpha_j)$ we know

$$f'(\alpha_i) = a \prod_{j \neq i} (\alpha_i - \alpha_j).$$

We get

$$\text{Res}(f, f') = a^{2n-1} \prod_{i=1}^n \prod_{j \neq i} (\alpha_i - \alpha_j) = (-1)^{\frac{n(n-1)}{2}} a^{2n-1} \prod_{i < j} (\alpha_i - \alpha_j)^2,$$

where each pair of $(i, j), i < j$ accounts for $-(\alpha_i - \alpha_j)^2$.

Remark $f = a_0 X^n + \dots + a_n, a = a_0$. Now

$$\text{Res}(f, f') = \begin{vmatrix} a & & \\ & \ddots & \\ na & & * \end{vmatrix} = a \cdot \text{poly}(a_0, \dots, a_n),$$

so the definition is reasonable when the leading coefficient is not 1.